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Simple ion-pairing reversed phase LC-UV method for the relative quantification of capping species in mRNA-based modalities

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ABSTRACT

Messenger RNA (mRNA)-based therapeutics and vaccines rely on proper 5′ capping to ensure translational efficiency, stability, and reduced reactogenicity. Current analytical approaches for capping evaluation often rely on mass spectrometry (MS) and fluorinated solvents, which are costly, technically demanding, and not always suitable for quality control (QC) laboratories. Here, we describe the development of a simple, robust, and QC-compatible ion-pairing reversed-phase liquid chromatography method coupled with UV detection (IP-RPLC-UV) for the relative quantification of key 5′ capping species in mRNA modalities, including uncapped, Cap 0, Cap 1, and Cap G transcripts, as well as non-templated additions (+G). The method employs targeted cleavage of the 5′ end using an oligo hybridization and RNase H digestion strategy, enabling baseline resolution of capping species within 20 min. A systematic screening of various ion-pairing reagents demonstrated that butylammonium acetate offered the optimal balance between retention and resolution, while a design of experiments approach identified critical parameters influencing performance and ensured method robustness. Method sensitivity was confirmed with a lower limit of quantification at 0.01 mg/mL for uncapped species, and repeatability testing showed consistent results across samples. This study introduces the first IP-RPLC-UV method tailored for routine relative quantification of cap structures, providing a practical alternative to MS-based workflows and facilitating the implementation of accessible, reliable analytical control strategies for mRNA drug development.

1. Introduction

Messenger RNA (mRNA)-based therapeutics and vaccines have emerged at the forefront of modern biopharmaceutical development, offering a versatile platform for vaccines, protein replacement therapies, and gene editing. Unlike traditional small molecules or monoclonal antibodies (mAb), mRNA modalities use the cell's own translational machinery to produce therapeutic proteins directly in situ, thus simplifying manufacturing and enabling rapid response to emerging pathogens.

To ensure safety, efficacy, and reproducibility, mRNA modalities must be evaluated against a set of Quality Attributes (QAs), including attributes that are i) intrinsic to the mRNA molecule (such as sequence integrity, length distribution, and modification pattern), ii) related to formulation characteristics (lipid nanoparticle composition, encapsulation efficiency, and particle size distribution), and iii) linked with stability parameters (e.g., integrity profile under storage conditions) [1–4]. Importantly, the QAs discussed here are introduced to frame the analytical relevance of the methods evaluated in this study and should not be interpreted as implying the existence of established, ICH-like regulatory validation frameworks or acceptance criteria for mRNA quality control. At present, comprehensive harmonized guidelines analogous to ICH Q2 for small molecules or biologics are not yet available for mRNA-based products. Accordingly, the present work focuses on QC-oriented analytical performance and methodological applicability, rather than on full regulatory-style method validation, which remains an evolving area for mRNA modalities.

Among them, the 5′ cap plays a pivotal role in translational efficiency, immunogenicity, and overall performance [5]. Its structure consists of a modified guanosine nucleotide linked via a 5′–5′

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triphosphate bridge to the 5′ end of the mRNA transcript [6]. It both protects the transcript from exonucleolytic degradation and serves as the recognition motif for the eukaryotic translation initiation factor complex (eIF4F), facilitating ribosome recruitment and efficient protein synthesis [7]. In the context of mRNA modalities, proper capping is indispensable: uncapped or improperly capped transcripts are rapidly degraded by cellular nucleases, trigger undesired innate immune responses, and yield suboptimal protein expression [8].

To ensure vaccines and therapeutics efficacy and safety, synthetic mRNA molecules are capped to mimic naturally occurring cap forms. The simplest, Cap G, comprises a guanosine (G) linked to the first nucleotide with a triphosphate bridge; Cap 0, contains a 7-methylguanosine (m^7G) instead of the first guanosine; and Cap 1 carries an additional 2′-*O*-methyl modification on the ribose of the first nucleotide. Cap methylation patterns (Cap 0 versus Cap 1) modulate immunogenicity, with higher methylation states more closely resembling native host mRNA and thus diminishing the activation of immune response [7]. In contrast to methylated cap structures, Cap G species lack N7-methylation and are therefore poorly recognized by cap-binding proteins such as eIF4E, resulting in reduced translational efficiency. From an analytical and quality control perspective, their quantification is important, as Cap G species can arise as by-products of in vitro transcription or incomplete capping and may impact overall product performance and batch-to-batch consistency [9]. Some uncapped species might also be encountered in the samples, showing mono-, di-, and triphosphate species without any guanosine attached [10]. It is noteworthy that polymerase non-templated addition can also result in the formation of alternative species, such as non-templated G addition at the 5′-end [10,11].

To generate capped species, synthetic mRNA manufacturing typically employs two main capping strategies: i) co-transcriptional capping, where cap analogues are incorporated during in vitro transcription (IVT), and ii) post-transcriptional enzymatic capping, which uses capping enzymes to modify the 5′ end of the IVT transcript [12,13]. Co-transcriptional capping uses cap analogs, such as anti-reverse cap analogues (ARCA) or CleanCap reagents, to enhance cap incorporation and ensure correct 5′ orientation [6,14]. Enzymatic capping, by contrast, employs vaccinia capping enzymes followed by specific 2′-*O*-methyltransferases to generate Cap 1 [5]. Each approach yields distinct capping species distributions: co-transcriptional capping can result in a mixture of uncapped, correctly capped, and mis-capped transcripts, whereas enzymatic methods often achieve near-quantitative capping, but require additional downstream purification steps to remove enzymes material and impurities generated by the digestion [5].

A variety of analytical techniques have been developed to characterize cap formation and methylation status. Cap-specific immunoassays employ mAb against m^7G or 2′-*O*-methylated cap structures, offering rapid, high-throughput screening capabilities [15]. However, they may lack absolute specificity, as some antibodies cross-react with internal m^7G residues or fail to distinguish between Cap 0 and Cap 1 species [16]. Other methods usually involve the specific digestion and liberation of 5′ cap and few attached nucleotides to ensure their detection. Indeed, the very small changes in physico-chemical properties induced by the addition of a single methyl group on a very large mRNA species is often insufficient to allow any clear separation and characterization. To this end, ribozyme digestion combined with polyacrylamide gel electrophoresis has been successfully used to characterize different capping species [17], but ion-pair reversed-phase liquid chromatography (IP-RPLC) is usually considered as the reference method for this application. In IP-RPLC, retention is governed by a combination of electrostatic interactions between charged analytes and ion-pairing reagents and the effective hydrophobic contribution of the ion-paired species at the stationary-phase surface, which makes the technique well suited for the analysis of hydrophilic nucleic acid-based molecules [18–20]. It is often combined with mass spectrometry (MS) allowing precise determination of cap structure and methylation pattern at the molecular level, and might also give information on the general sequence of the mRNA strand [21]. The commonly used methods to specifically remove the capping species are uridine specific digestion with RapiZyme MC1 [22], RNAse H digested probe [10,23–25], pistol ribozyme [26], or digestion of a DNA probe [27]. The IP-RPLC method usually involved mobile phases composed of diisopropylethylamine and hexafluoro-2-propanol (DIPEA-HFIP) [22,26–28], triethylamine (TEA) and HFIP [10,23], or dibutylamine (DBA) and HFIP [25] in combination with MS. However, this setup requires sophisticated instrumentation, specialized technical expertise, and MS coupled with IP-RPLC can be difficult. Finally, hydrophilic liquid chromatography (HILIC) has also been explored to improve coupling to MS for capping evaluation, but its use remains very limited [24].

To facilitate the analysis of 5′ cap populations present in mRNA-based modalities, we developed a simple IP-RPLC-UV method to relatively quantify the four main capping species (i.e. uncapped, Cap 0, Cap 1, Cap G) present in different mRNA samples, including unpurified research-stage material. The goal was to establish a routine-compatible and cost-effective method that does not require the use of MS, nor perfluorinated reagent (HFIP). The use of a non-MS detection strategy is driven by routine QC considerations, where methods must be robust, cost-effective, and broadly deployable across quality control laboratories, including environments in which access to MS is limited or impractical. The method is based on the specific cleavage of oligo hybridized 5′-end with RNAse H and achieves baseline separation of all the species in a 20-min run. Sample preparation was optimized to ensure adequate sensitivity and removal of digestion-related impurities. A design of experiments (DoE) approach was implemented to identify critical parameters influencing response variability and to evaluate method robustness under varying operational conditions. Finally, repeatability and sensitivity were assessed using research-stage mRNA-LNP drug products. To our knowledge, this is the first IP-RPLC-UV QC-compatible method developed for the analysis of mRNA capping species.

2. Experimental

2.1. Materials and reagents

RNase-free water, acetic acid (AA, ≥ 99.8%), butylamine (BA, 99.5%), diisopropylethylamine (DIPEA, ≥ 99%), hexylamine (HA, ≥ 99%), and pentafluoropropionic acid (PFPA, ≥ 97%) were all purchased from Sigma-Aldrich (Buchs, Switzerland). LC-MS grade acetonitrile and methanol were purchased from Thermo Fischer Scientific (Reinach, Switzerland). A Milli-Q water purification system from Millipore (Bedford, MA, USA) was used to prepare HPLC mobile phases. DNA LoBind tubes of 0.5 mL purchased from Eppendorf (Hamburg, Germany) were used to incubate the samples in an Incubator Thermomixer Confort from Eppendorf. Polypropylene 100 µL vial inserts were purchased from Macherey-Nagel (Duren, Germany).

Design of experiments were performed on JMP 17.0 1 software from JMP Statistical Discovery (Vary, NC, USA). Statistical analyses were performed on GraphPad Prism 10.4.2.(GraphPad Software, USA). Chromatograms were produced using color blind friendly themes.

2.2. Samples

Samples were provided by the Sanofi mRNA Center of Excellence and are described in Table S1. They consisted of mRNA species with significant cap variation, specifically selected to challenge the analytical method development. The proportions of capping species were determined using a reference MS method [29].

2.3. Sample preparation

Samples underwent cleavage of capping species using an oligo hybrid and RNAse H digestion. For this purpose, samples were first

prepared at a concentration of 2.0 mg/mL in LoBind Eppendorf tubes. Hybridization was performed by adding 5 µL of oligo-hybrid to each sample, followed by two incubations: 10 min at 75 °C and 10 min at 23 °C. The resulting RNA-DNA hybrid was specifically cleaved by adding 3 µL RNAse H 5 U/µL, 1 µL RNAse H buffer 10×, and 3 µL shrimp alkaline phosphatase 1 U/µL, followed by an incubation during 40 min at 37 °C under gentle agitation. Finally, 1 µL 0.5 M ethylenediaminetetraacetic acid was added to chelate bivalent ions introduced by the buffers. Samples were then transferred to LC vials with polypropylene insert and injected.

2.4. UHPLC instrumentation, columns and experimental conditions

The chromatographic measurements were performed on an ACQUITY UPLC H-class system (Waters, Milford, MA, USA), equipped with a 10 µl flow through needle injector, a quaternary solvent manager (QSM) equipped with a 380 µl mixing chamber, a column manager, and an ACQUITY photodiode array detector. UV detection was set at a wavelength of 260 nm. The column was an ACQUITY BEH C18 130 Å, 1.7 µm, 2.1 × 100 mm. Data acquisition and instrument control were performed by Empower 3 software (Waters). Resolution (Rs) was calculated using peak widths at half height with the following Eq. (1).

$$Rs = \frac{1.18(t_{R2} - t_{R1})}{(W_2 + W_1)} \quad (1)$$

where t_{R1} and t_{R2} are the retention times of peaks 1 and 2, respectively, and $W_2 + W_1$ the sum of peak widths at 50% peak height, and with the assumption of Gaussian symmetric peaks [30].

The mobile phases were always composed of an equimolar concentration of ion-pairing reagent (IPR) and acetic acid. Concentrations between 100 and 150 mM were evaluated. The pH was adjusted by adding small amount of acetic acid. In this study, hexafluoro-2-propanol (HFIP) was not considered as an alternative to acetic acid, since HFIP is typically used to boost electrospray ionization-MS sensitivity, which was not the goal of the present work. To screen the IPRs, a linear gradient from 3 to 50% B at a flow rate of 0.4 mL/min was systematically used.

The final method used an injection volume of 9 µL, a flow rate of 0.4 mL/min, and a column temperature of 80 °C. mobile phase A (MPA) was 125 mM BA, 125 mM acetic acid, pH 7.0 in pure water. Mobile phase B (MPB) was 125 mM BA, 125 mM acetic acid, 75% acetonitrile (MeCN), 25% H_2O. The same extra amount of acetic acid used to adjust the pH in MPA was also added to MPB. Strong and weak washes were MeCN:H_2O 50:50 and 100% MeCN, respectively. The optimized gradient is described in Table S2.

As a good practice, when changing the nature of IPR, we used a dedicated wash to remove IPR from the system, as previously reported [31]. It consisted of a 45-min wash at 65 °C at 0.2 mL/min using a 1:1:1:1 mixture of isopropanol, methanol, acetonitrile, and water, with 0.1% pentafluoropropionic acid, followed by a 10-min wash with 100% acetonitrile.

2.5. Cap identification by UHPLC-MS

Mass spectrometry was used to confirm the identification of the peaks obtained in IP-RPLC-UV. Mobile phases were identical to the one used in IP-RPLC-UV, even if the concentrations of ammonium acetate and IPR were not optimal for MS sensitivity and induced a lot of adducts. The loss of sensitivity was considered acceptable because identification was still possible. Acquisition was performed on a Waters Bioaccord Premier instrument, a high-resolution compact benchtop time-of-flight mass spectrometry (TOF MS) system, equipped with a UV detector. Table S3 summarizes the ions used for MS identification.

3. Results and discussion

3.1. Capping species and oligo-hybrid design

As shown in Fig. 1, four main capping species were expected, namely Uncapped, Cap 0, Cap G, and Cap 1. To obtain them, a hybrid oligonucleotide (HO) was designed, consisting of RNA and DNA bases complementary to the 5'-UTR end of the mRNA sequence of interest. During sample preparation, the mRNA was first linearized at 75 °C, followed by gradual cooling to room temperature to enable efficient mRNA:HO hybridization. RNAse H, an enzyme that specifically recognizes double strands of RNA:DNA, only cleaves after the fifth base of the hybrid. The proportion of the four capping species obtained was influenced by the efficacy of the capping reaction during the manufacturing process.

3.2. Selection of the most suitable ion-pairing reagents

In ion-pair reversed-phase liquid chromatography, analyte retention is generally rationalized using complementary mechanistic models, including mobile-phase ion-pair formation, dynamic adsorption of the ion-pairing reagent (IPR) at the stationary-phase surface, and electrostatic interaction models. While the relative contribution of these mechanisms remains system-dependent, their common implication for method development is that both the chemical nature and effective surface activity of the IPR govern retention and selectivity, beyond simple hydrophobicity considerations [18–20]. In this context, selecting an appropriate IPR for the intended application is critical, as the different capping species must be sufficiently retained and clearly separated for effective UV detection. Importantly, if retention is dominated by the hydrophobicity of the IPR, the subtle hydrophobic differences introduced by single methyl substitutions on the capping species may be masked, leading to a loss of chromatographic resolution.

Triethylammonium acetate buffer (TEAA) is commonly used as an additive for such analyses and is recommended by official guidelines for the characterization of 5′ capping species [32]. Recent studies on IPR selection for oligonucleotides [33] and large mRNA molecules [31] have demonstrated a direct link between oligonucleotide retention and IPR hydrophobicity. Therefore, to optimize the separation and resolution of the 5′ capping species, we also investigated moderately hydrophobic IPR (diisopropylethylammonium acetate (DIPEAA) and butylammonium acetate (BAA)), and more hydrophobic IPR (hexylammonium acetate (HAA)). Based on prior work, we selected 100 mM as an optimal concentration to ensure sufficient retention, and 75% MeCN in mobile phase B to support solubility and elution [31]. The column temperature was maintained at 80 °C to maximize kinetic performance and improve peak shape. For peak identification, selected samples containing only one capping species were injected under these conditions.

In initial experiments, important retention shifts were observed regardless of the IPR used. Since peak identification relied on retention times, minimizing variability was essential. Increasing the column equilibration time from 20 to 40 column volumes yielded only slight improvement. However, replacing the 100 µL mixed chamber with a larger 380 µL chamber significantly stabilized retention times, reducing CVs from 5.5% to just 0.1%, as shown in Fig. 2. This underscores the importance of adequate mixing of mobile phase components under IP-RPLC conditions, especially when using a quaternary pumping system.

Surprisingly, adequate retention of the capping species could not be achieved under TEAA conditions using the widely used BEH C18 column. The capping species were insufficiently retained (elution near the column dead volume) and coeluted with impurities originating from the sample preparation process. Given that our probe design generates relatively small 5′ capping species composed of only five nucleotides bearing a relatively high number of negative charges due to the triphosphate group, we hypothesized that TEAA lacked sufficient hydrophobicity to retain them effectively. Moreover, selectivity between the different species was clearly inadequate. We believe this outcome is

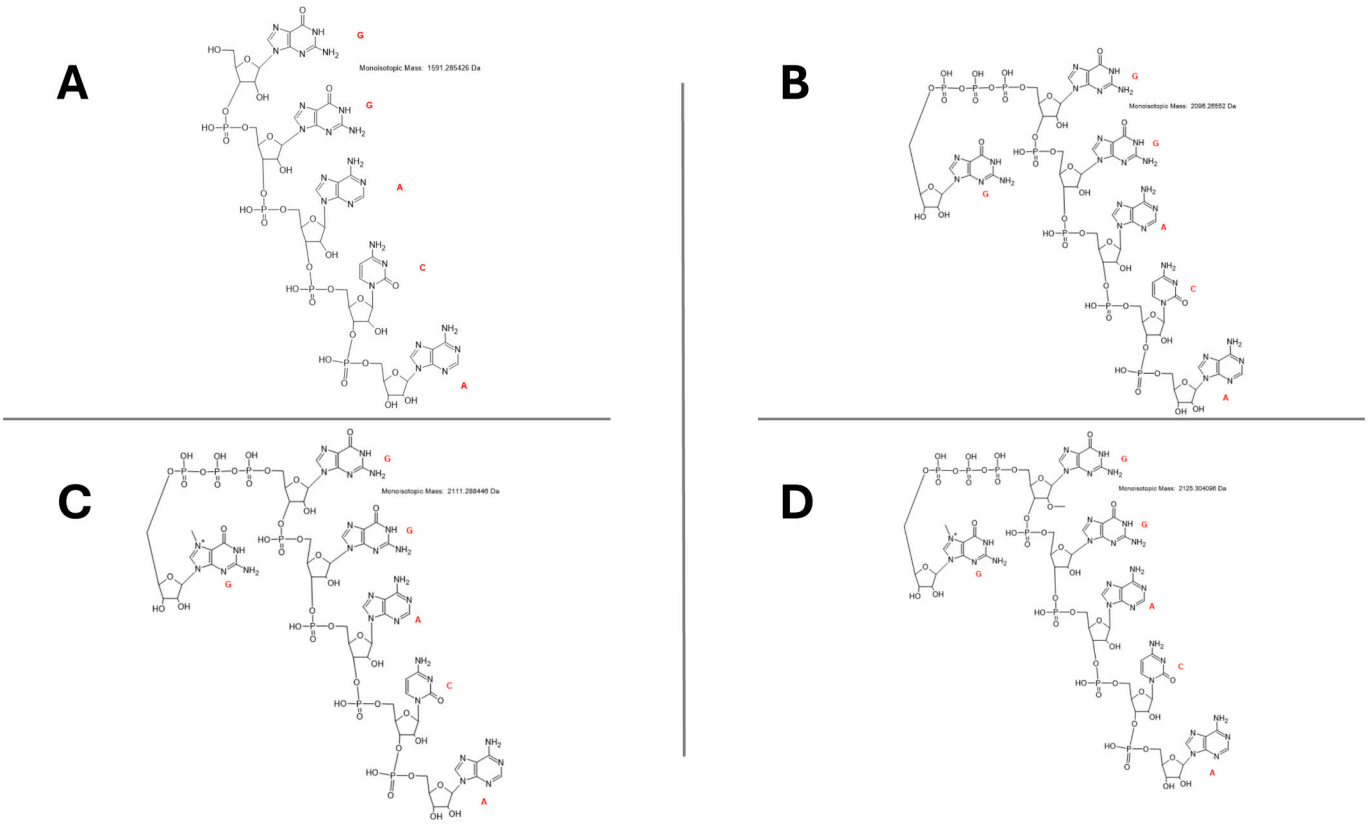


Fig. 1. Four main 5′-capping species structures, obtained after oligo-hybridization and RNAse H digestion. A) Uncapped; B) Cap 0; C) Cap G; D) Cap 1.

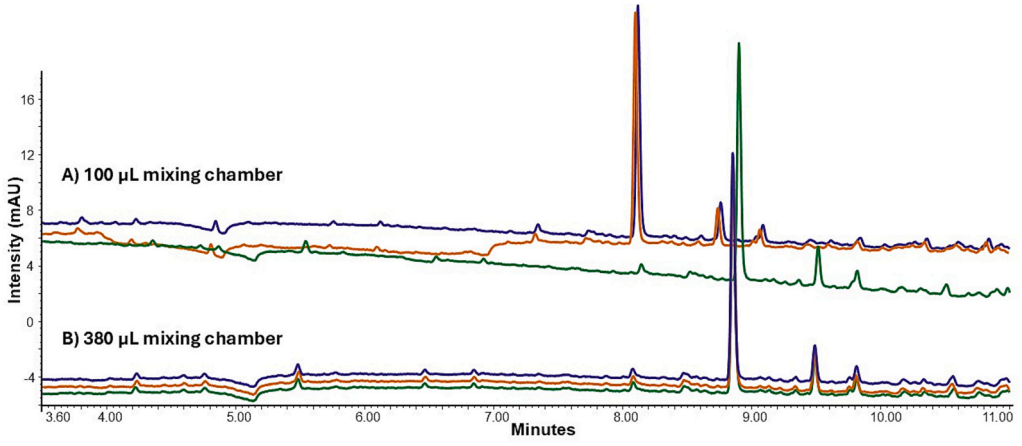


Fig. 2. Chromatograms obtained for a sample analyzed 3 times with different mixing chambers. A) the mixing chamber has a volume of 100 µL. B) the mixing chamber has a volume of 380 µL. Important retention time shifts were observed when 100 µL mixing chamber was used. Mobile phase A (MPA) was 100 mM butylammonium acetate in pure water. Mobile phase B (MPB) was 100 mM butylammonium acetate in 75% MeCN, 25% H2O. Analyses were performed with a linear gradient from 3 to 50% MPB at a temperature of 80 °C and a flow rate of 0.4 mL/min.

strongly influenced by the length of the 5′ capping species, and retention might be improved by changing the oligo-hybrid and increasing the number of nucleotides attached to the 5′ G. However, extending the number of oligonucleotides may compromise selectivity between closely related capping species.

In the present study, we explored several more hydrophobic ion-pairing reagents. Among them, HAA provided an excellent retention of all the 5′ capping species (see Fig. 3). However, retention was entirely governed by the high hydrophobicity of the IPR, making it impossible to resolve species differing by only a single methyl group (cap 1 vs. cap 0). Attempts to fine-tune gradient conditions to improve separation were inconclusive.

Based on the suboptimal retention and/or selectivity observed with TEAA and HAA, we investigated moderately hydrophobic IPRs, such as DIPEAA and BAA, aiming to achieve an optimal balance between retention of all 5′-capping species and selectivity between species differing by a single methyl group. DIPEAA was first tested and provided clear separation of Cap 0 and Cap 1 species, as shown in Fig. S1. However, this IPR consistently caused significant baseline drift at 260 nm, which may interfere with detection and integration of 5′ capping species. To address this issue, butylammonium acetate, an IPR with moderate hydrophobicity, was evaluated as an alternative. A clear improvement in

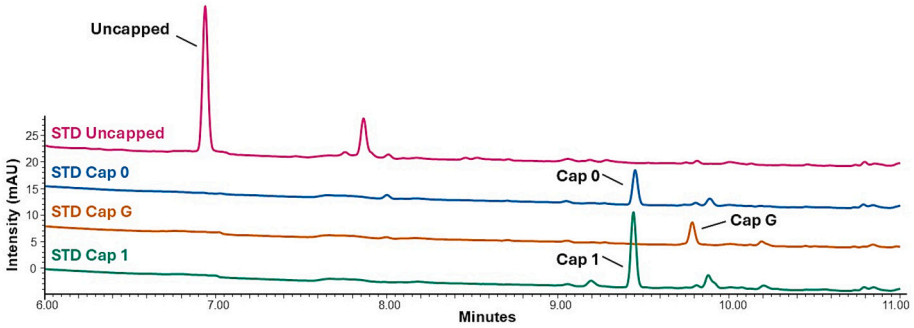


Fig. 3. Chromatograms obtained for the analysis of various samples, each containing only one type of capping species using hexylammonium acetate (HAA) in mobile phase. The initial part of the chromatograms (before 6 min) was not shown because they contained a lot of early-eluting impurities coming from the sample preparation and digestion. Mobile phase A: 100 mM HAA in pure water. Mobile phase B: 100 mM HAA in 75% MeCN. Column Acquity BEH C18 130 Å, 1.7 µm, 2.1 × 100 mm. Analyses were performed with a linear gradient from 3 to 50% MPB at a temperature of 80 °C and a flow rate of 0.4 mL/min.

baseline stability was observed, and good separation of all 5′-capping species was achieved, as shown in Fig. 4.

3.3. Optimization of the chromatographic conditions

Results obtained from non-purified samples showed more than the four expected capping peaks. MS was therefore used to identify these additional species, which were + G variants resulting from non-templated additions by RNA polymerase. We hypothesized that this phenomenon occurs because RNA polymerase has an intrinsic bias toward initiating transcription with guanosine triphosphate (GTP) [34]. Unlike bacterial RNA polymerases, which can initiate transcription with either adenosine triphosphate (ATP) or GTP, SP6 RNA polymerase strictly requires GTP as the initiating nucleotide at the 5′ end. As reported in early mechanistic studies, this high affinity for GTP leads to repeated incorporation events at the transcription start site, particularly when the DNA template contains cytosine-rich regions opposite the initiation site. Consequently, non-templated additions at the 5′ end produce transcripts carrying one or more additional guanosine residues, a phenomenon that has been previously reported in the literature [10].

To ensure baseline separation of the four main species and their respective +G variants while maximizing sensitivity, chromatographic conditions were optimized. Initially, gradient conditions were adjusted to achieve sufficient selectivity between all the 5′-capping species. A rather complex gradient program, varying from 5% to 11.5% B over 17.1 min was established, as detailed in Table S2. Next, a Design of Experiments (DoE) approach was employed to identify and understand the impact of additional parameters on resolution, sensitivity, and relative quantification performance. Following a risk assessment and preliminary experiments, three parameters were selected for investigation: injection volume (5–9 µL), BAA concentration (100–150 mM), and mobile phase pH (6–8). A two-level screening design was used to represent the upper and lower bounds of each parameter, as illustrated in Table 1. The responses evaluated included: Cap 1 relative percentage, signal-to-noise ratio (S/N) for the uncapped species, resolution 1 (between Cap 0 + G and Cap G), and resolution 2 (between Cap G and Cap 1 + G). These two resolution metrics were prioritized because the corresponding species were the most challenging to separate.

Chromatograms obtained through the DoE are detailed in Fig. S2. For Cap 1 Relative %, none of the tested parameters showed a significant impact (all *P*-values >0.05, $R^2 = 0.51$), indicating that there was no meaningful effect on the quantitative accuracy of the method when varying injection volume, IPR concentration, or mobile phase pH. Moreover, the tested conditions yielded similar Cap 1 relative % values, with an RSD of only 0.7% across all conditions, showing robustness of this response. However, the low R^2 value (0.51) indicates that the model explains only part of the observed variability, suggesting that other unidentified factors might influence this response. Despite the low coefficient of variation, which confirms the method's suitability for the intended application, an additional experimental design was carried out (in section 3.4) to verify that all relevant parameters were considered. Conversely, the S/N of the uncap response was significantly influenced by the three factors: injection volume (*P*-value = 0.00240, $R^2 = 1$), pH (P-value = 0.00264, $R^2 = 1$), and BAA concentration (P-value = 0.00590, $R^2 = 1$). Specifically, S/N increased with higher injection volume, higher pH, and lower BAA concentration, confirming that method sensitivity can be improved by tuning these parameters. Finally, the combined Resolution 1 and 2 analysis identified pH as the only significant factor (P-value = 0.00046, $R^2 = 0.94$), with pH 7 providing an optimal balance between the two resolutions. In contrast, pH 6 and 8 resulted in resolution values below 1.5 for Resolution 2 and Resolution 1, respectively. The impact of the different factors on the four responses

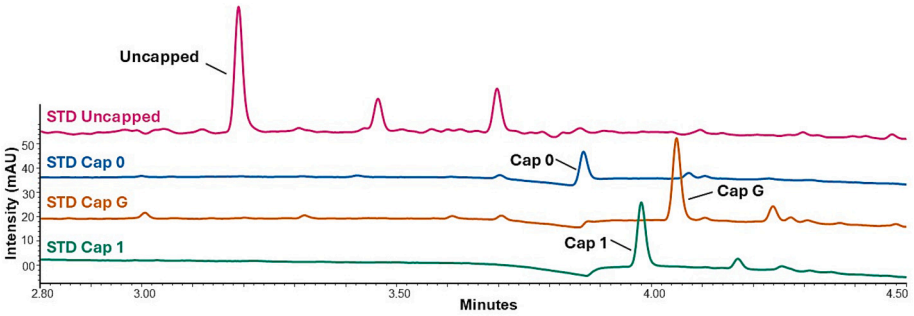


Fig. 4. Chromatograms obtained for the analysis of various samples, each containing only one type of capping species using butylammonium acetate (BAA) in the mobile phase. The initial part of the chromatograms (before 2.8 min) was not shown because they contained a lot of early-eluting impurities coming from the sample preparation and digestion. Mobile phase A: 100 mM BAA in pure water. Mobile phase B: 100 mM BAA in 75% MeCN. Column Acquity BEH C18 130 Å, 1.7 µm, 2.1 × 100 mm. Analyses were performed with a linear gradient from 3 to 50% MPB at a temperature of 80 °C and a flow rate of 0.4 mL/min.

Table 1
Factors evaluated in the first screening DoE for method optimization and robustness evaluation. The screened parameters were injection volume (5–9 µL), pH (7–9), and butylammonium acetate (BAA) concentration (100–150 mM).

Configuration	Injection Volume (µL)	BAA (mM) in MP	pH in MP	Cap 1 relative percentage	S/N uncap	Resolution 1	Resolution 2
+++	9	150	8	86	8.4	0	6.9
+−−	9	100	6	86.3	7.7	7.5	1.1
0	7	125	7	85.4	7.1	4	4.9
−−+	5	100	8	85.7	7.6	0	6.3
0	7	125	7	84.7	6.9	4	5.2
− + −	5	150	6	85	4.1	7.7	1.2

can be visualized in Fig. S3. Based on these observations, the following optimal conditions were selected: injection volume = 9 µL, pH = 7, and BAA concentration = 125 mM. Temperature was set at 80 °C because species were still sufficiently retained, and high temperatures are known to improve mass transfer resistance by reducing mobile phase viscosity, leading to higher diffusion coefficient, and improved peak shapes. Under these conditions, the four main species and their non-templated additions were clearly separated, and sensitivity was found to be fit-for-purpose, as shown in Fig. 5.

3.4. Evaluation of method robustness

Next, we conducted a follow-up screening with a fractional factorial DoE to assess the impact of small variations in experimental conditions, comparable to those likely to occur during the routine method use, on Cap 1 relative percentage. In this case, five parameters were investigated, namely column temperature, injection volume, BAA concentration, percentage of MeCN in mobile phase B, and mobile phase pH, using the conditions described in Table 2. The significance of the effect was evaluated using a confidence interval of 0.95.

The impact of the different factors is shown in Fig. S4. This experimental design produced a significantly improved model (compared to section 3.3.) for Cap 1 relative %, with $R^2 = 1$, indicating excellent predictive capability. Within the experimental domain, all factors showed statistically detectable effects ($p < 0.05$), but with minimal practical impact: the standard deviation across all DOE results was only 0.8%, which falls well below our practical relevance threshold. Therefore, despite the statistical significance of these factors, the method demonstrates robust performance with acceptable variation across the parameter ranges studied. This comprehensive analysis confirms that the method is suitable for routine use, as small variations in critical parameters do not meaningfully impact the quantitative determination of Cap 1 relative percentage.

Table 2
Factors evaluated in the factorial fractional design for robustness evaluation. The screened parameters were column temperature (78–82 °C), injection volume (8–10 µL), butylammonium acetate (BAA) concentration in mobile phases (123–127 mM), acetonitrile (MeCN) percentage in mobile phase B (74–76%), and pH (6.8–7.2).

Levels	Column temperature (°C)	Injection volume (µL)	BAA concentration (mM)	% MeCN	pH
−++−+	78	10	127	74	7.2
+−+−−	82	8	127	74	6.8
−−++−	78	8	127	76	6.8
+++++	82	10	127	76	7.2
+−−++	82	8	123	76	7.2
0	80	9	125	75	7
−−−−+	78	8	123	74	7.2
−+−+−	78	10	123	76	6.8
0	80	9	125	75	7
++−−−	82	10	123	74	6.8

3.5. Qualification of the method

Capping species are obtained by hybridizing a DNA probe to the 5′ end, followed by digestion of the resulting RNA:DNA duplex with RNase H. Obtaining samples with defined capping proportions is particularly challenging, and no standards were available. Nevertheless, we performed a comprehensive qualification of the analytical method, assessing key performance characteristics, including limits of quantification, repeatability, intermediate precision, linearity, and accuracy.

The lower limit of quantification (LLOQ) was evaluated using the uncapped species because it contains only five nucleotides, whereas the

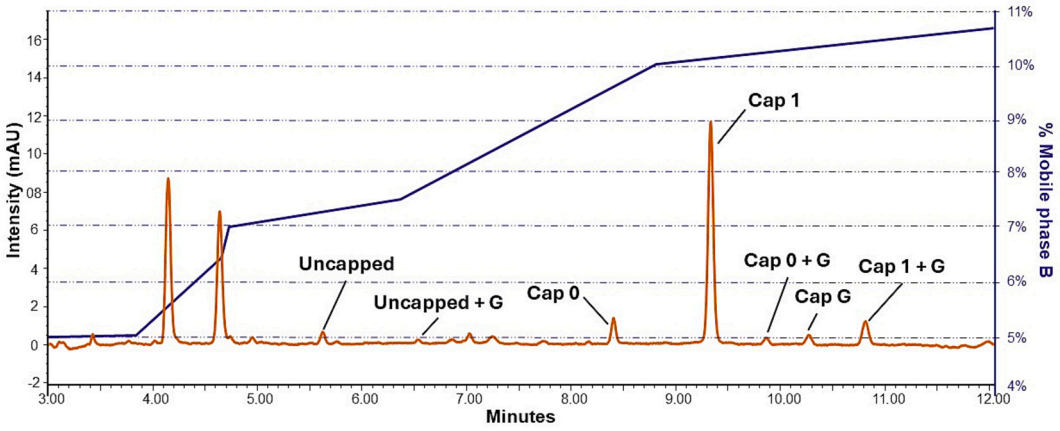


Fig. 5. Chromatograms obtained for the analysis of a non-purified sample using the optimized conditions and gradient. The four main capping species and their +G non-templated additions were identified by MS. The initial part of the chromatograms (before 3 min) was not shown because they contained a lot of early-eluting impurities coming from the sample preparation and digestion. Mobile phase A: 125 mM BAA in pure water, pH 7, and injection volume of 9 µL. Mobile phase B: 125 mM BAA in 75% MeCN. Column Acquity BEH C18 130 Å, 1.7 µm, 2.1 × 100 mm. Analyses were performed with the optimized gradient described in Table S2 at a temperature of 80 °C and a flow rate of 0.4 mL/min.

other species contain six nucleotides, meaning the uncapped species is expected to produce a lower UV response at 260 nm. The LLOQ corresponded to the lowest detected concentration of uncapped species that generated a signal-to-noise ratio of 10. A sample containing only uncapped mRNA was subjected to oligo-hybridization and treated with RNase H, yielding a final concentration of 1.3 mg mRNA/mL. This sample was further diluted in RNase-free water to obtain the lowest concentration maintaining a S/N > 10. At 0.013 mg/mL, the S/N of the uncapped species was 18 (see Fig. S5). Based on these results, the LLOQ was fixed at 0.01 mg/mL, and it is likely that the LLOQs of the other capping species with additional nucleotides are higher.

Because the different capping species vary in nucleotides length, we compared their signals obtained by UV and MS. The purpose of this comparison was to determine whether UV data required correction using species-specific response factors. The MS method is described in Materials and Methods. We found that all response factors ranged between 0.8 and 1.2. Based on this limited variability, and following a fit-for-purpose analytical approach informed by general validation principles (such as those described in ICH Q2(R2)), a response factor of 1 was applied for all species.

Repeatability and intermediate precision were assessed for relative Cap 1% in four series of four replicates, performed with two different analysts using different columns and HPLC instruments on two separate days. The obtained CV were 0.2% for repeatability (16 samples) and 0.4% for intermediate precision (32 samples). Method linearity was assessed using Cap 1 relative percentage at five different levels (0%, 25%, 50%, 75%, and 100% DS). The calibration curve was linear within the investigated range with the following regression equation: $y = 0.991\times - 0.148$, and $r^2 = 0.999$. Accuracy was evaluated at three levels (0%, 50%, and 100% DS), with experimental values of 0%, 48.3%, and 99.6%, respectively.

3.6. Application to real samples

Finally, we applied the developed IP-RPLC-UV method on six samples with different capping proportions and compared the results obtained with the UHPLC-MS reference method. It is noteworthy that Cap 0 was excluded from this evaluation, because none of the samples contained detectable levels of this species. This reflects the high capping efficiency of the production process, which yields Cap 0 levels below the detection limit, even when using the UHPLC-MS reference method. At such trace or non-detectable levels, no adverse impact on product performance is expected, and further increases in analytical sensitivity are therefore not required for the intended quality control application.

Correlation analysis showed excellent agreement between the two methods for Cap 1 and Uncapped species ($R^2 > 0.99$), with only minor deviations from the line of identity (Fig. 6). For Cap G, a strong correlation was also observed ($R^2 = 0.81$), although larger deviations from the line of identity were noted. These deviations were primarily associated with samples in which Cap G levels approached the UV detection limit, resulting in increased relative uncertainty. These minor discrepancies between UV- and MS-based measurements observed at trace levels near the quantification limit do not affect the intended QC interpretation of overall capping quality.

Method agreement was further evaluated using Bland-Altman analysis. Following confirmation of normality of the MS-UV differences by the Shapiro-Wilk test, mean biases of −1.61, 1.21, and 0.43 percentage points were observed for Cap 1, Uncapped, and Cap G species, respectively. These biases are small relative to the typical abundance of the corresponding species and do not affect the qualitative ranking or QC

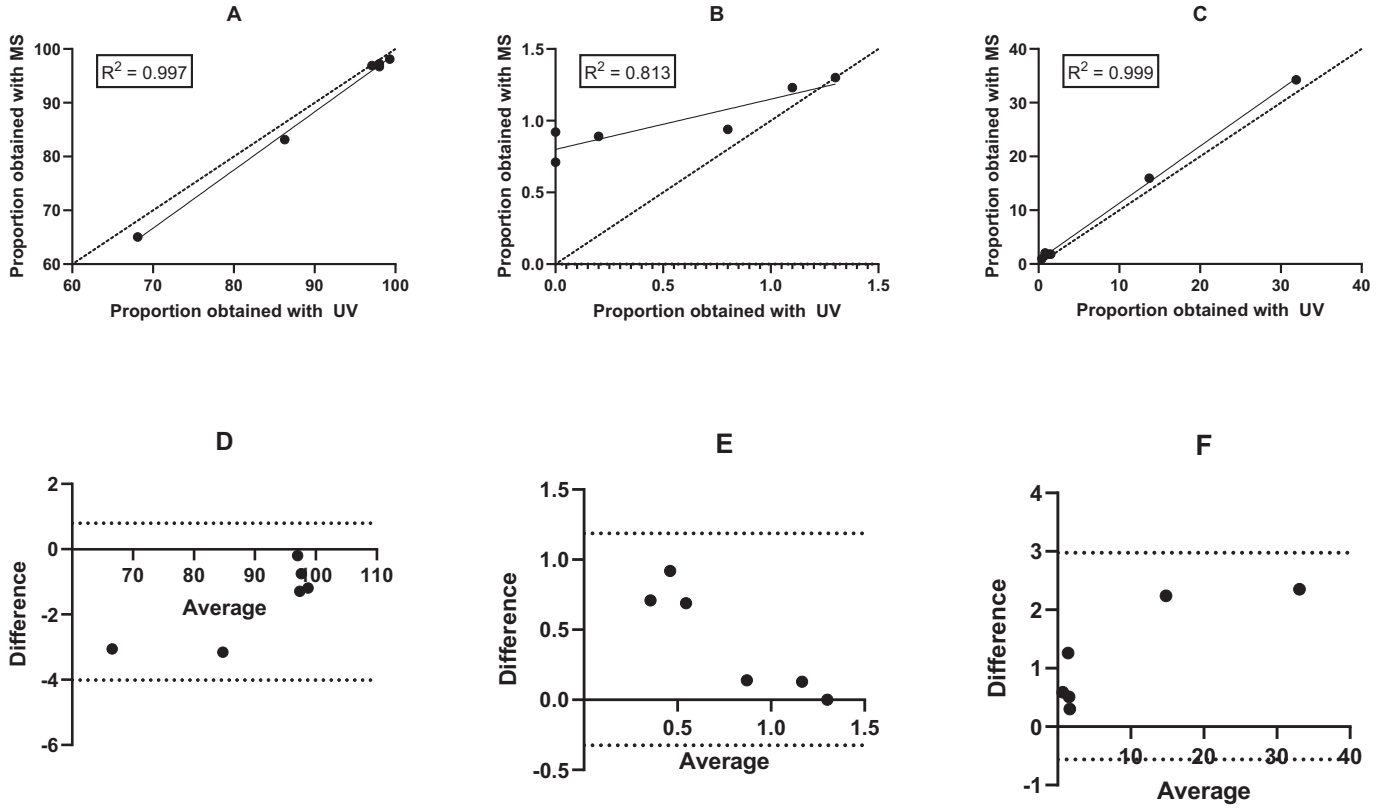


Fig. 6. Comparison of capping species quantification obtained by LC–MS and LC–UV detection. Six different mRNA samples exhibiting varying capping proportions were analyzed with both detectors. Subfigures A, B, and C are scatterplots generated for each detected species (%Cap 1, %Cap G, and %Uncapped, respectively). In each scatterplot, the dotted line represents the line of identity (y = x), and the solid line represents the correlation. Subfigures D, E, and F are Bland-Altman plots generated for each detected species (%Cap 1, %Cap G, and %Uncapped, respectively). Cap 0 was excluded from the analysis since none of the tested samples contained detectable amounts of this species.

interpretation of the samples. Given the intended use of the method for routine QC monitoring and the low abundance of Cap G in the tested samples, the level of agreement between UV and MS was considered acceptable.

4. Conclusion

In this work, we established a robust and QC-compatible IP-RPLC-UV method for the relative quantification of capping species in mRNA therapeutics and vaccines. This approach represents an advancement in analytical methodology by eliminating the need for sophisticated MS instrumentation while maintaining suitable analytical performance. The sample preparation integrates oligo-hybridization with RNase H digestion to achieve specific cleavage of capped species.

A systematic screening of four ion-pairing reagents (IPR) covering a broad range of hydrophobicity revealed that moderately hydrophobic IPRs, such as butylammonium acetate (BAA), offered optimal performance. BAA ensured appropriate retention of small capped species, while enabling the separation of species differing by only a single methyl group. Our results highlight that the use of IPRs with limited hydrophobicity is critical when sequence differences are targeted.

After optimization of the chromatographic separation using a screening DoE, the following conditions were identified as optimal: 80 °C, pH 7, 125 mM BAA, V_{inj} = 9 µL, multistep gradient from 5 to 11.5% B, with B being a mixture of 75:25 MeCN:water. The method's robustness was confirmed through an additional DoE approach. The LLOQ for uncapped species was determined at 0.01 mg/mL, highlighting the high sensitivity of the method.

Beyond its analytical performance, our method provides valuable insights into the molecular heterogeneity of mRNA modalities. The optimized workflow was successfully applied to non-purified, research-grade mRNA samples produced in-house, which exhibited varied distributions of capping species. It enabled baseline resolution of the four main 5′-capping forms (uncapped, Cap 0, Cap G, and Cap 1) as well as +G species arising from polymerase non-templated additions. The identification and characterization of these +G species further underscore the complexity of mRNA synthesis and the need for comprehensive analytical characterization. Consistent results were obtained between UV and MS detection, demonstrating the reliability of the approach.

Overall, the method presented here is straightforward, transferable, and well-suited for routine QC analysis, offering robustness, sensitivity, and repeatability. The method avoids reliance on complex instrumentation such as MS, ensuring easy implementation in QC-environments. While MS remains essential for comprehensive characterization of complex capping profiles, our method offers a complementary approach that balances analytical performance with practical implementation considerations. As mRNA modalities continue to advance into clinical applications, accessible analytical methods will be critical for ensuring consistent quality and accelerating research and development timelines.

CRediT authorship contribution statement

Jonathan Maurer: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Denis Bouchard:** Writing – review & editing, Investigation, Formal analysis. **Agathe Bousquier:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Camille Malburet:** Writing – review & editing, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Jean-François Cotte:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Davy Guillarme:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: This work was partially funded by Sanofi. Denis Bouchard, Agathe Bousquier, Camille Malburet, and Jean-François Cotte are Sanofi employees and hold shares and/or stock options in the company. Jonathan Maurer and Davy Guillarme declare no competing interests

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jchromb.2026.124922.

Data availability

The data that has been used is confidential.

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